Khadijah Aamir Chaudhry

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LinkedIn · GitHub · Portfolio

EDUCATION

Lahore University of Management Sciences (LUMS) 2023–2027
BSc Computer Science · Minor in Computer Engineering

Lahore Grammar School 2018–2023
A Levels: 4 As · O Levels: 9 A*s · SAT: 1530

RESEARCH INTERESTS

Secure embedded systems · Hardware roots of trust · Hardware–software co-design · Trusted execution · Applied cryptography

RESEARCH EXPERIENCE

Research Assistant, CHASE Lab, LUMS Aug 2025–present
Supervisor: Dr. Muhammad Ali Siddiqi

- To study runtime hardware Trojan detection, reviewed the research area and built a simulated Rocket Chip and Verilator testbed that captures RISC-V transaction traces.
- To compare post-quantum methods on constrained devices, benchmarked Kyber-512, Light-SABER, and Dilithium-II on ESP32 for latency, memory, network cost, and estimated energy.

Research Assistant, SysNet Lab, LUMS Jan–May 2026
Supervisor: Dr. Naveed Anwar Bhatti

- To explore accessible eyesight screening, co-developed EyeSense's smartphone time-of-flight phase-map and machine-learning pipeline for refractive-error measurement.
- To check the approach against real measurements, supported data collection across lab and eye-camp settings. The team studied 120 participants and reported 1.15 D mean absolute error.

SELECTED PROJECTS

On Chip Behavioral Monitoring · early draft 2026–present

- To prepare traces for runtime monitoring research, configured four RV32 Rocket Chip variants and captured TileLink/MMIO activity with Verilator.
- To compare ordinary and anomalous execution, replayed benign and trigger-active firmware traces for later detector evaluation.

Protected Media Containers · early draft 2026–present

- With mobile operating systems able to access ordinary media files, researched a design that keeps camera and audio plaintext inside trusted execution.
- Specified encrypted storage, display, and sharing flows so users can retain policy controls while using an untrusted phone operating system.

EyeSense: Mobile Eyesight Screening · project write-up Jan–May 2026

- To investigate low-cost refractive-error screening, co-developed a phone-based ToF sensing and ML pipeline that measures phase-map changes.
- The team evaluated the method with 120 participants and reported 1.15 D mean absolute error, exploring a path to easier preliminary eyesight checks.

Post-Quantum Cryptography on ESP32 · project write-up · code Aug–Dec 2025

- To understand post-quantum cryptography on small devices, implemented and benchmarked Kyber-512, Light-SABER, and Dilithium-II on ESP32.
- Compared latency, memory, network cost, and estimated energy to show practical trade-offs for embedded deployments.

AasPass Peer Tutoring · app demo

- For a student–tutor matching app, led Android authentication and administrator features, including LUMS email verification and profile setup.

- Added moderation controls to help administrators manage a safer peer tutoring community.

Spotify Mood Classification and Recommendation · write-up

- To explore how audio patterns relate to listening moods, used audio features, PCA, and K-means to group songs.
- Built mood classification and similarity-based recommendations to help listeners find music with a similar feel.

ROM-Based Digital Lock · project report

- To build a simple physical access-control demonstration, implemented an EPROM keypad lock using state-machine logic.
- Added an unlock indicator and incorrect-entry alarm, making accepted and rejected codes clear to the user.

TEACHING EXPERIENCE

Teaching Assistant, CS100, LUMS Aug 2026–present

- To support about 450 first-year students learning programming, designed and conducted labs and prepared and graded quizzes.
- Held weekly office hours and invigilated exams, giving students regular access to help and assessment support.

PROFESSIONAL EXPERIENCE

Marketing Manager, Fintelligent Solutions Jul 2026–present

- To establish a consistent public identity for the consultancy, coordinated team delivery of brand identity and office collateral.
- Coordinated LinkedIn launch content and the company website so prospective clients can learn about services and make contact.

Data Collection Coach, Uplift AI Jul–Aug 2024

- To support collection of a Pakistani-language AI dataset, led 10 data collectors and organized field outreach.
- Ran daily quality reviews and feedback sessions to help improve consistency in collected examples.

LEADERSHIP & SERVICE

Peer Ambassador Leader, LUMS Aug 2026–present

- To help 15 incoming students settle into university, led orientation and introductions to faculty and campus culture.
- Continue monthly support sessions through students' first year to provide a familiar point of contact.

Women in STEM Coordinator, LUMS SPADES Sep 2024–May 2025

- To make STEM programming more welcoming to women, helped launch SPADES' first women-only STEM event at PSIFI.
- The event brought together more than 100 participants for a dedicated space to engage with science and technology.

Co-Founder and Logistics Director, Umeed-e-Nau NGO Sep 2021–May 2023

- To create inclusive learning opportunities, co-founded an NGO serving underprivileged and disabled children.
- Organized educational activities and visits that gave children opportunities to learn and participate together.

SELECTED GRADUATE COURSEWORK

Internet of Things · Applied Cryptography · Topics in Computer and Network Security · Distributed Systems

HONORS AND AWARDS

- 100% Merit Scholarship, Lahore Grammar School · Aug 2021–May 2023
- Shortlisted writing submission, LUMSx Nexus · May 2025
- Honorable Mention, EducationUSA Pakistan Writing Contest · Sep 2022

TECHNICAL SKILLS

Programming languages: C · C++ · Python · JavaScript
Embedded and hardware: ESP32 · RISC-V · Rocket Chip · Verilator · FreeRTOS
Machine learning and sensing: ToF sensing · CNNs · ridge regression
Tools and methods: Docker · Git · post-quantum cryptography benchmarking